BEFORE CALCULUS

CALCULUS:

It is the study of continuous change. Isaac Newton and Gottfried Wilhelm Leibniz the pioner of Calculus. It deals with limits, differentiation and integration of functions of one or more variables. The two major concepts of Calculus are

Derivatives: It gives us the rate of change of a function. It describes the function at a particular point.

Integrals: it gives us the area under curve. It gathers the different values of a function over number of values.

Analytical Geometry: It is also Known as coordinate geometry or Cartesian geometry, is the study of geometry using a coordinate System. It is a combination of algebra and geometry used to model geometric objects Rene Descartes (Renatius Cartesius in Latin) is the founder/father of it.

Functions: A function f is a rule that gives unique output with denoted by X with each input. Input is denoted by x and output is denoted by f(x) by unique it means "exactly one".

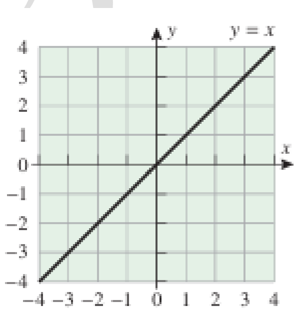
Independent and Dependent Variables:

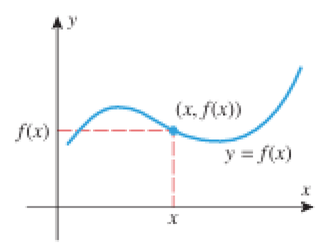
$$y=f(x)$$

Here the variable x is called the independent variable (or argument) of f and the variable y is called the dependent variable of f.

GRAPHS OF FUNCTIONS:

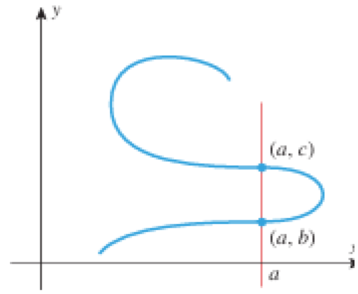
If f is a real-valued function of a real variable, then the graph of f in the xy-plane is defined to be the graph of the equation y = f(x). For example, the graph of the function f(x) = x is the graph of the equation y = x.





▲ **Figure 0.1.5** The y-coordinate of a point on the graph of $y = f(x)$ is the value of f at the corresponding x-coordinate.

THE VERTICAL LINE TEST: A curve in the xy-plane is the graph of some function f if and only if no vertical line intersects the curve more than once.



▲ **Figure 0.1.7** This curve cannot be the graph of a function.

Absolute Value: The absolute value of a number or integer is the actual distance of the integer from zero, in a number line. Therefore, the absolute value is always a positive.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

0.1.4 PROPERTIES OF ABSOLUTE VALUE *If a and b are real numbers, then*

(*a*)	$\lvert -a\rvert = \lvert a\rvert$	A number and its negative have the same absolute value.
(*b*)	$\lvert ab\rvert = \lvert a\rvert\,\lvert b\rvert$	The absolute value of a product is the product of the absolute values.
(*c*)	$\lvert a/b\rvert = \lvert a\rvert/\lvert b\rvert,\ b \neq 0$	The absolute value of a ratio is the ratio of the absolute values.
(*d*)	$\lvert a+b\rvert \leq \lvert a\rvert + \lvert b\rvert$	The ***triangle inequality***

A piecewise function: it is a function that has different parts or pieces. It means it has different definitions depending upon the value of the input.

Verify (1) by using a graphing utility to show that the equations $y = \sqrt{x^2}$ and $y = |x|$ have the same graph.

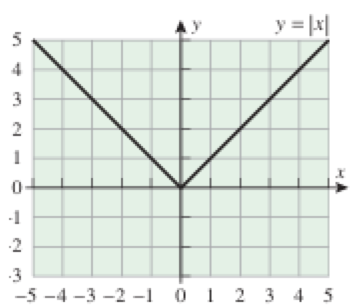


Figure 0.1.9

DOMAIN AND RANGE:

If x and y are related by the equation y = f(x), then the set of all allowable inputs (x-values) is called the domain of f, and the set of outputs (y-values) that result when x varies over the domain is called the range of f.

Also; If a real-valued function of a real variable is defined by a formula, and if no domain is stated explicitly, then it is to be understood that the domain consists of all real numbers for which the formula yields a real value. This is called the natural domain of the function.

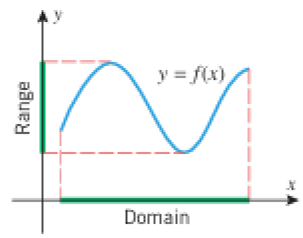


Figure 0.1.12 The projection of $y = f(x)$ on the x-axis is the set of allowable x-values for f, and the projection on the y-axis is the set of corresponding y-values.

ARITHMETIC OPERATIONS ON FUNCTIONS:

0.2.1 DEFINITION Given functions f and g, we define

$$(f+g)(x) = f(x) + g(x)$$
$$(f-g)(x) = f(x) - g(x)$$
$$(fg)(x) = f(x)g(x)$$
$$(f/g)(x) = f(x)/g(x)$$

For the functions $f+g$, $f-g$, and fg we define the domain to be the intersection of the domains of f and g, and for the function f/g we define the domain to be the intersection of the domains of f and g but with the points where $g(x) = 0$ excluded (to avoid division by zero).

COMPOSITION OF FUNCTIONS:

Given functions f and g, the composition of f with g, denoted by $f \circ g$, is the function defined by

$$(f \circ g)(x) = f(g(x))\ f(g(x))$$

The domain of $f \circ g$ is defined to consist of all x in the domain of g for which g(x) is in the domain of f.

EXPRESSING AFUNCTION AS A COMPOSITION:

Many problems in mathematics are solved by "decomposing" functions into compositions of simpler functions. For example, consider the function h given by

$$h(x) = (x+1)^2$$

To evaluate h(x) for a given value of x, we would first compute x + 1 and then square the result. These two operations are performed by the functions g(x) = x +1 and f(x)=x^2. We can express h in terms of f and g by writing

$$h(x) = (x+1)^2 = [g(x)]^2 = f(g(x))$$

so we have succeeded in expressing h as the composition h = f ∘g. The thought process in this example suggests a general procedure for decomposing a function h into a composition h = f ∘g:

• Think about how you would evaluate h(x) for a specific value of x, trying to break the evaluation into two steps performed in succession.

• The first operation in the evaluation will determine a function g and the second a function f.

• The formula for h can then be written as h(x) = f(g(x)). For descriptive purposes, we will refer to g as the “inside function” and f as the “outside function” in the expression f(g(x)). The inside function performs the first operation and the outside function performs the second.

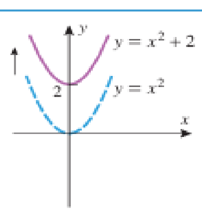
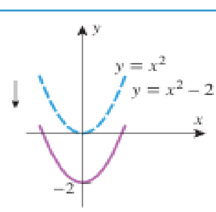
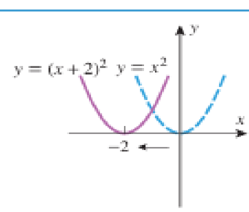
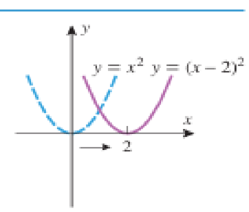
COMPOSING FUNCTIONS

FUNCTION	$g(x)$ INSIDE	$f(x)$ OUTSIDE	COMPOSITION
$(x^2+1)^{10}$	x^2+1	x^{10}	$(x^2+1)^{10} = f(g(x))$
$\sin^3 x$	$\sin x$	x^3	$\sin^3 x = f(g(x))$
$\tan(x^5)$	x^5	$\tan x$	$\tan(x^5) = f(g(x))$
$\sqrt{4-3x}$	$4-3x$	$\sqrt{x}$	$\sqrt{4-3x} = f(g(x))$
$8+\sqrt{x}$	$\sqrt{x}$	$8+x$	$8+\sqrt{x} = f(g(x))$
$\frac{1}{x+1}$	$x+1$	$\frac{1}{x}$	$\frac{1}{x+1} = f(g(x))$

Graphical Changes:

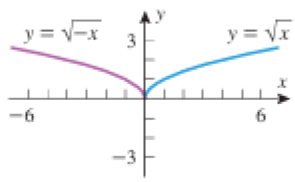
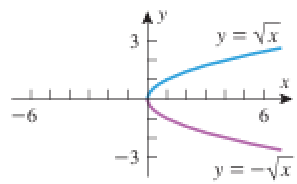
TRANSLATIONS: The geometric effect on the graph of y = f(x) o fading or subtracting a positive constant c to f or to its independent variable x.

TRANSLATION PRINCIPLES

OPERATION ON $y = f(x)$	Add a positive constant c to $f(x)$	Subtract a positive constant c from $f(x)$	Add a positive constant c to x	Subtract a positive constant c from x
NEW EQUATION	$y = f(x) + c$	$y = f(x) - c$	$y = f(x + c)$	$y = f(x - c)$
GEOMETRIC EFFECT	Translates the graph of $y = f(x)$ up c units	Translates the graph of $y = f(x)$ down c units	Translates the graph of $y = f(x)$ left c units	Translates the graph of $y = f(x)$ right c units
EXAMPLE				

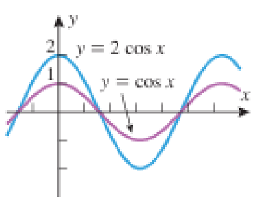
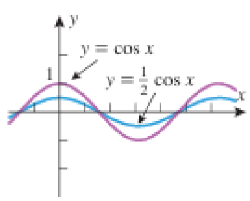
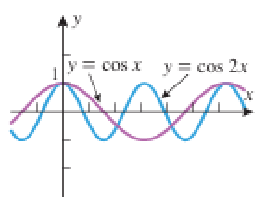
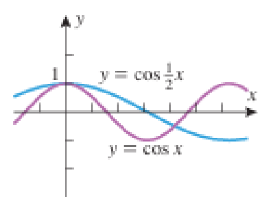
REFLECTIONS: The graph of y= f(−x) is the reflection of the graph of y = f(x) about they-axis because the point (x, y) on the graph of f(x) is replaced by (−x, y). Similarly, the graph of y=−f(x) is the reflection of the graph of y=f(x)about the x-axis because the point (x, y) on the graph of f(x) is replaced by (x, −y). The equation y=−f(x)is equivalent to−y= f(x).

REFLECTION PRINCIPLES

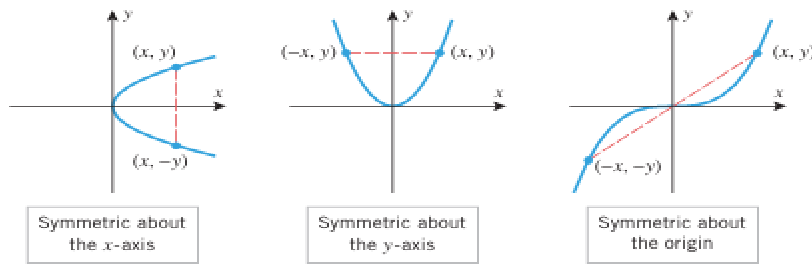
OPERATION ON $y = f(x)$	Replace x by $-x$	Multiply $f(x)$ by -1
NEW EQUATION	$y = f(-x)$	$y = -f(x)$
GEOMETRIC EFFECT	Reflects the graph of $y = f(x)$ about the y-axis	Reflects the graph of $y = f(x)$ about the x-axis
EXAMPLE		

STRETCHES AND COMPRESSIONS: Multiplying f(x) by a positive constant c has the geometric effect of stretching the graph of y=f(x) in they-direction by a factor of c if c >1 and compressing it in the y direction by a factor of 1/c if 0 < c < 1.

STRETCHING AND COMPRESSING PRINCIPLES

OPERATION ON $y = f(x)$	Multiply $f(x)$ by c $(c > 1)$	Multiply $f(x)$ by c $(0 < c < 1)$	Multiply x by c $(c > 1)$	Multiply x by c $(0 < c < 1)$
NEW EQUATION	$y = cf(x)$	$y = cf(x)$	$y = f(cx)$	$y = f(cx)$
GEOMETRIC EFFECT	Stretches the graph of $y = f(x)$ vertically by a factor of c	Compresses the graph of $y = f(x)$ vertically by a factor of $1/c$	Compresses the graph of $y = f(x)$ horizontally by a factor of c	Stretches the graph of $y = f(x)$ horizontally by a factor of $1/c$
EXAMPLE				

SYMMETRY: There are three types of symmetries: symmetry about the x-axis, symmetry about the y-axis, and symmetry about the origin. As illustrated in the figure, a curve is symmetric about the x-axis if for each point (x, y) on the graph the point (x, −y) is also on the graph, and it is symmetric about the y-axis if for each point (x, y) on the graph the point (−x, y) is also on the graph. A curve is symmetric about the origin if for each point (x, y) on the graph, the point (−x, −y) is also on the graph.



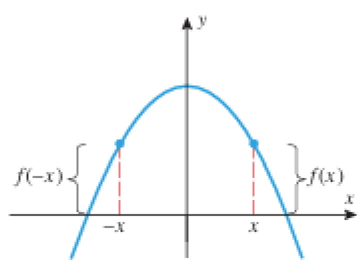
0.2.3 THEOREM (*Symmetry Tests*)

(*a*) *A plane curve is symmetric about the y-axis if and only if replacing x by $-x$ in its equation produces an equivalent equation.*

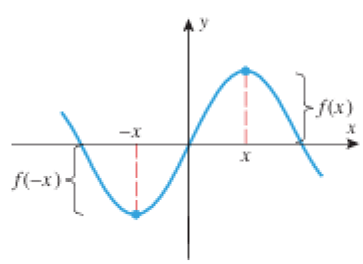
(*b*) *A plane curve is symmetric about the x-axis if and only if replacing y by $-y$ in its equation produces an equivalent equation.*

(*c*) *A plane curve is symmetric about the origin if and only if replacing both x by $-x$ and y by $-y$ in its equation produces an equivalent equation.*

EVEN AND ODD FUNCTIONS: A function f is said to be an even function if f(−x) = f(x) and is said to be an odd function if f(−x) = −f(x). Geometrically, the graphs of even functions are symmetric about the y-axis because replacing x by−x in the equation y = f(x) yields y = f(−x), which is equivalent to the original equation y=f(x) Similarly, it follows from that graphs of odd functions are symmetric about the origin. Some examples of even functions are x^2, x^4, x^6, and cosx; and some examples of odd functions are x^3, x^5, x^7, and sinx.

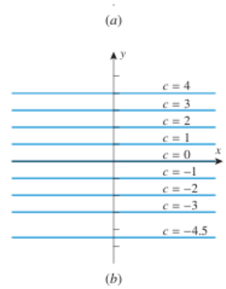


▲ **Figure 0.2.9** This is the graph of an even function since $f(-x) = f(x)$.



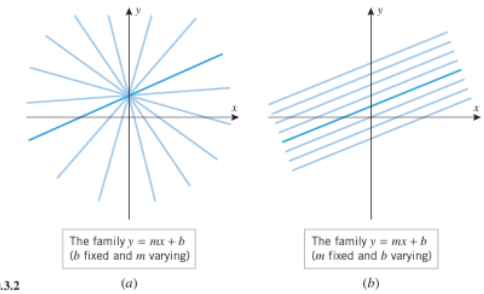
▲ **Figure 0.2.10** This is the graph of an odd function since $f(-x) = -f(x)$.

FAMILIES OF CURVES: The graph of a constant function f(x)= c is the graph of the equation y = c, which is the horizontal line shown in Figure 0.3.1a. If we vary c, then we obtain a set or family of horizontal lines such as those in Figure 0.3.1b. Constants that are varied to produce families of curves are called parameters. For example, recall that an equation of the form y = mx + b represents a line of slope m and y-intercept b. If we keep b fixed and treat m as a parameter, then we obtain a family of lines whose members all have y-intercept b (Figure 0.3.2a), and if we keep m fixed and treat b as a parameter, we obtain a family of parallel lines whose members all have slope m.



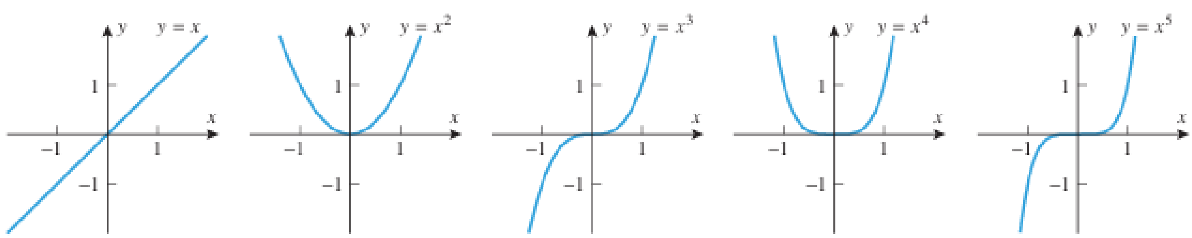
▲ Figure 0.3.1

▶ Figure 0.3.2



POWER FUNCTIONS:

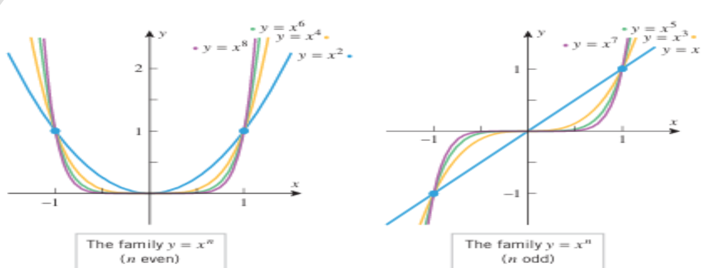
THE FAMILY $y = x^n$ A function of the form $f(x)= x^p$, where p is constant, is called a power function. For the moment, let us consider the case where p is a positive integer, say p = n. The graphs of the curves $y = X^n$ for n = 1, 2, 3, 4, and 5 are shown. The first graph is the line with slope 1 that passes through the origin, and the second is a parabola that opens up and has its vertex at the origin.



For $n \geq 2$ the shape of the curve $y = x^n$ depends on whether n is even or odd:

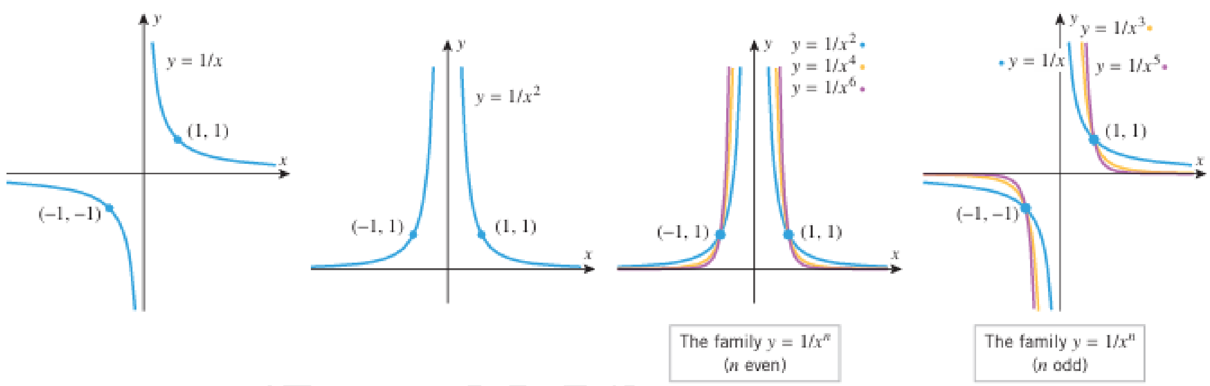
• For even values of n, the functions $f(x)= x^n$ are even, so their graphs are symmetric about the y-axis. The graphs all have the general shape of the graph of $y = x^2$, and each graph passes through the points (−1,1), (0,0), and (1,1). As n increases, the graphs become flatter over the interval $-1 < x < 1$ and steeper over the interval $x > 1$ and $x < -1$.

• For odd values of n, the functions $f(x)= x^n$ are odd, so their graphs are symmetric about the origin. The graphs all have the general shape of the curve $y = x^3$, and each graph passes through the points (−1, −1), (0,0), and (1,1). As n increases, the graphs become flatter over the interval $-1 < x < 1$ and steeper over the interval $x > 1$ and $x < -1$.



THE FAMILY $y = x^{-n}$. If p is a negative integer, say $p = -n$, then the power functions $f(x) = x^p$ have the form $f(x) = x^{-n} = 1/x^n$. Figure shows the graphs of $y = 1/x$ and $y = 1/x^2$. The graph of $y = 1/x$ is called an equilateral hyperbola. The shape of the curve $y = 1/x^n$ depends on whether n is even or odd:

- For even values of n, the functions $f(x) = 1/x^n$ are even, so their graphs are symmetric about the y-axis. The graphs all have the general shape of the curve $y = 1/x^2$, and each graph passes through the points $(-1,1)$ and $(1,1)$. As n increases, the graphs become steeper over the intervals $-1 < x < 1$ and $0 < x < 1$ and flatter over the interval $x > 1$ and $x < -1$.
- For odd values of n, the functions $f(x) = 1/x^n$ are odd, so their graphs are symmetric about the origin. The graphs all have the general shape of the curve $y = 1/x$, and each graph passes through the points $(1,1)$ and $(-1, -1)$. As n increases, the graphs become steeper over the intervals $-1 < x < x$ 1 and $0 < x < 1$ and flatter over the interval $x > 1$ and $x < -1$.
- For both even and odd values of n the graph $y = 1/x^n$ has a break at the origin (called a discontinuity), which occurs because division by zero is undefined.



INVERSE PROPORTIONS:

Recall that a variable y is said to be inversely proportional to a variable x if there is a positive constant k, called the constant of proportionality, such that

$$y = k/x$$

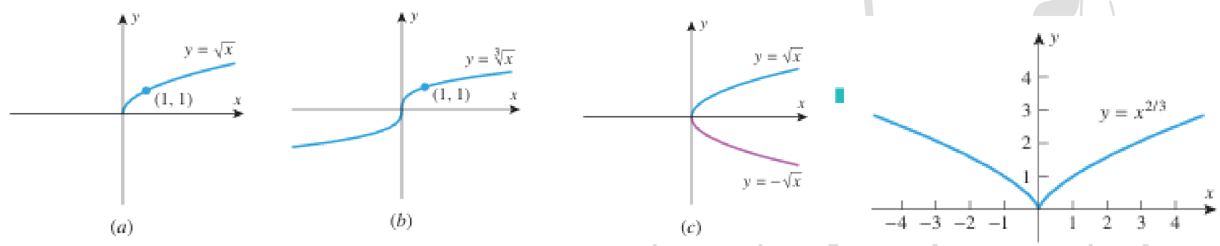
Since k is assumed to be positive, the graph of (1) has the same shape as $y = 1/x$ but is compressed or stretched in the y-direction. Also, it should be evident from equation (1) that doubling x multiplies y by 1/2, tripling x multiplies y by 1/3, and so forth. Equation (1) can be expressed as

$$xy = k,$$

which tells us that the product of inversely proportional variables is a positive constant. This is a useful form for identifying inverse proportionality in experimental data.

POWER FUNCTIONS WITH NONINTEGER EXPONENTS:

If $p = 1/n$, where n is a positive integer, then the power functions $f(x)= x^p$ have the form $f(x)= x^{1/n} = n\sqrt{x}$ in particular, if $n = 2$, then $f(x)= \sqrt{x}$, and if $n = 3$, then $f(x)= {}^3\sqrt{x}$. The graphs of these functions are shown in parts (a) and (b) of Figure. Since every real number has a real cube root, the domain of the function $f(x)= 3\sqrt{x}$ is $(-\infty, +\infty)$, and hence the graph of $y = 3\sqrt{x}$ extends over the entire x-axis. In contrast, the graph of $y = \sqrt{x}$ extends only over the interval $[0, +\infty)$ because $\sqrt{x}$ is imaginary for negative x. As illustrated in, the graphs of $y = \sqrt{x}$ and $y =-\sqrt{x}$ form the upper and lower halves of the parabola $x = y^2$. In general, the graph of $y = n\sqrt{x}$ extends over the entire x-axis if n is odd, but extends only over the interval $[0, +\infty)$ if n is even. Power functions can have other fractional exponents. Some examples are $f(x)= x^{2/3}$, $f(x) = {}^5\sqrt{x^3}$, $f(x) = x^{-7/8}$.



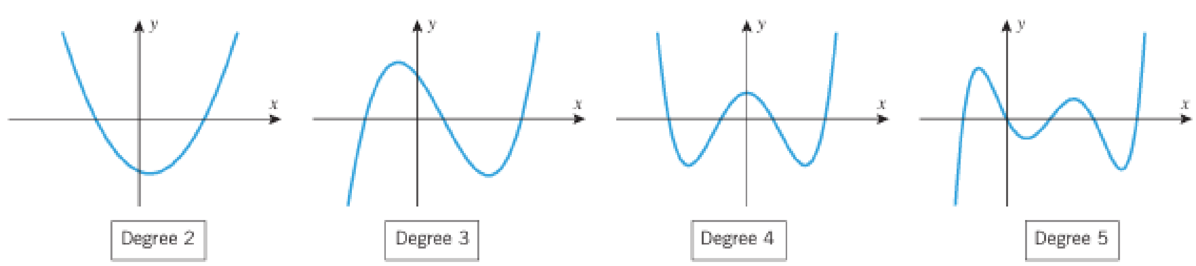
POLYNOMIALS: A polynomial in x Is a function that is expressible as a sum of finitely many terms of the form cx^n, where c is a constant and n is a nonnegative integer. The function $(x^2 - a)^3$ is also a polynomial because it can be expanded by the binomial formula (see the inside front cover) and expressed as a sum of terms of the form cx^n . A general polynomial can be written in either of the following forms, depending on whether one wants the powers of x in ascending or descending order:

$$c_0 + c_1x + c_2x_2 + \cdots + c_nx_n$$

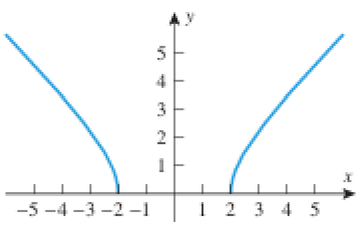
$$c_nx_n + c_{n-1}x_{n-1} + \cdots + c_1x + c_0$$

The constants $c_0, c_1, \ldots, c_n$ are called the coefficients of the polynomial. When a polynomial is expressed in one of these forms, the highest power of x that occurs with a nonzero coefficient is called the degree of the polynomial. Nonzero constant polynomials are considered to have degree 0, since we can write $c = cx_0$. Polynomials of degree 1, 2, 3, 4, and 5 are described as linear, quadratic, cubic, quartic, and quintic, respectively.

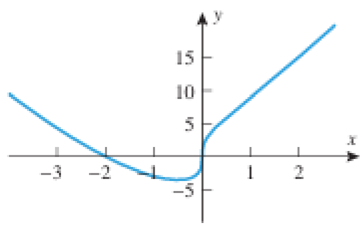
The natural domain of a polynomial in x is $(-\infty, +\infty)$,since the only operations involved are multiplication and addition; the range depends on the particular polynomial.



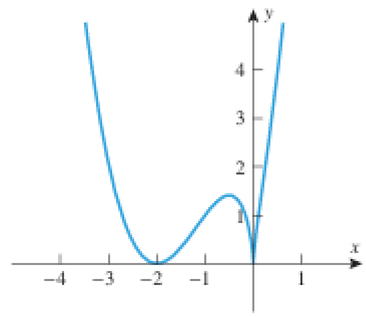
ALGEBRAIC FUNCTIONS: Functions that can be constructed from polynomials by applying finitely many algebraic operations (addition, subtraction, multiplication, division, and root extraction) are called algebraic functions.



$y = \sqrt{x^2 - 4}$



$y = 3\sqrt[3]{x}\,(2 + x)$



$y = x^{2/3}(x + 2)^2$

THE FAMILIES:

y =AsinBx and y =AcosBx

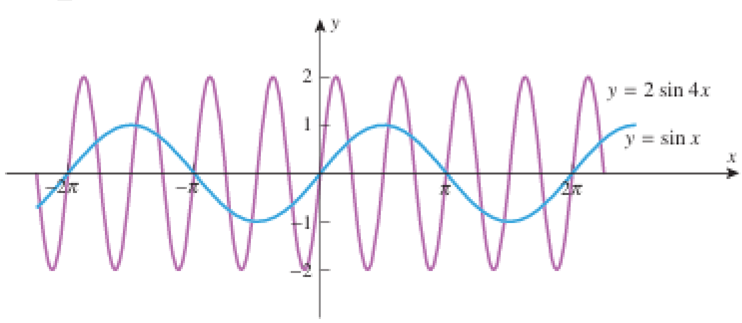
In this text we will assume that the in dependent variable of a trigonometric function is in radians unless otherwise stated. Many important applications lead to trigonometric functions of the form

f(x)= Asin(Bx −C) and g(x)= Acos(Bx −C)

where A, B, and C are nonzero constants. The graphs of such functions can be obtained by stretching, compressing, translating, and reflecting the graphs of

y = sinx and y = cosx

If A and B are positive, then the effect of the constant A is to stretch or compress the graphs of y = sinx and y = cosx vertically and the effect of the constant B is to compress or stretch the graphs of sinx and cosx horizontally. For example, the graph of y = 2sin4x can be obtained by stretching the graph of y = sinx vertically by a factor of 2 and compressing it horizontally by a factor of 4. (Recall that the multiplier of x stretches when it is less than 1 and compresses when it is greater than 1.) Thus, as shown in Figure, the graph of y = 2sin4x varies between −2and 2, and repeats every 2π/4 = π/2 units.



In general, if A and B are positive numbers, then the graphs of $y = A\sin Bx$ and $y = A\cos Bx$ oscillate between $-A$ and A and repeat every $2\pi/B$ unit, so we say that these functions have amplitude A and period $2\pi/B$. In addition, we define the frequency of these functions to be the reciprocal of the period, that is, the frequency is $B/(2\pi)$. If A or B is negative, then these constants cause reflections of the graphs about the axes as well as compressing or stretching them; and in this case the amplitude, period, and frequency are given by

$$\text{amplitude} = |A|,\ \text{period} = 2\pi/|B|,\ \text{frequency} = |B|/2\pi$$

THE FAMILIES: $y = A\sin(Bx - C)$ and $y = A\cos(Bx - C)$

To investigate the graphs of the more general families

$$y = A\sin(Bx - C) \text{ and } y = A\cos(Bx - C)$$

it will be helpful to rewrite these equations as

$$y = A\sin B(x - C/B) \text{ and } y = A\cos B(x - C/B)$$

In this form we see that the graphs of these equations can be obtained by translating the graphs of

$$y = A\sin Bx \text{ and } y = A\cos Bx$$

to the left or right, depending on the sign of C/B. For example, if $C/B > 0$, then the graph of

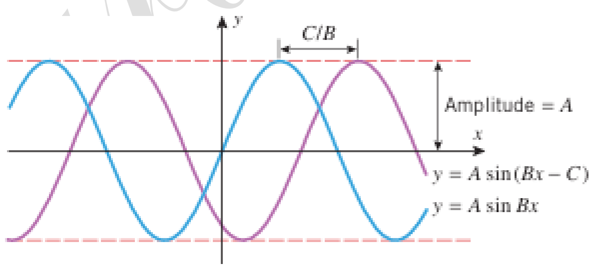
$$y = A\sin[B(x - C/B)] = A\sin(Bx - C)$$

can be obtained by translating the graph of $y = A\sin Bx$ to the right by C/B units.

if $C/B < 0$, then the graph of

$$y = A\sin[B(x - C/B)] = A\sin(Bx - C)$$

can be obtained by translating the graph of $y = A\sin Bx$ to the left by $|C/B|$ units.



INVERSE FUNCTIONS: The idea of solving an equation $y=f(x)$ for x as a function of y, say $x=g(y)$, is one of the most important ideas in mathematics. Sometimes, solving an equation is a simple process; for example, using basic algebra the equation:

$$y=x^3+1 \rightarrow y = f(x)$$

can be solved for x as a function of

$$x= y^3-1 \rightarrow x=g(y)$$

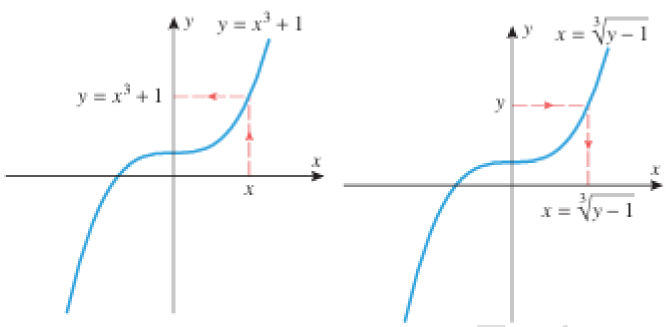
The first equation is better for computing y if x is known, and the second is better for computing x if y is known.

Our primary interest in this section is to identify relationships that may exist between the functions f and g when an equation y=f(x) is expressed as x=g(y), or conversely. For example, consider the functions $f(x)=x^3+1$ and $g(y)= \sqrt[3]{y-1}$ discussed above. When these functions are composed in either order, they cancel out the effect of one another in the sense that

$$g(f(x)) = \sqrt[3]{f(x) - 1} = \sqrt[3]{x^3 + 1 \ - 1} = x$$

$$(g(y)) = [g(y)]^3+1 = \sqrt[3]{(y-1)^3}+1 = y$$

Pairs of functions with these two properties are so important that there is special terminology for them.



0.4.1 DEFINITION If the functions f and g satisfy the two conditions

$$g(f(x)) = x \text{ for every } x \text{ in the domain of } f$$
$$f(g(y)) = y \text{ for every } y \text{ in the domain of } g$$

then we say that ***f is an inverse of g*** and ***g is an inverse of f*** or that ***f and g are inverse functions***.

and we can express the equations in Definition 0.4.1 as

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in the domain of } f$$
$$f(f^{-1}(y)) = y \quad \text{for every } y \text{ in the domain of } f^{-1}$$

We will call these the ***cancellation equations*** for f and f^{-1}.

CHANGING THE INDEPENDENT VARIABLE:

The above formulas use x as the independent variable for f and y as the independent variable for f^{-1}. Although it is often convenient to use different independent variables for f and f^{-1}, there will be occasions on which it is desirable to use the same independent variable for both. For example, if we want to graph the functions f and f^{-1} together in the same xy-coordinate system, then we would want to use x as the independent variable and y as the dependent variable for both functions. Thus, to graph the functions

$$f(x)= x^3 + 1 \text{ and } f^{-1}(y) = \sqrt[3]{y-1}$$

of the above Example in the same xy-coordinate system, we would change the independent variable y to x, use y as the dependent variable for both functions, and graph the equations

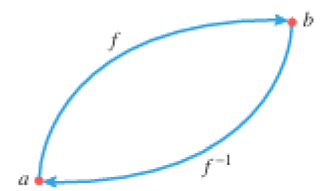
$$y = x^3+1 \text{ and } y = \sqrt[3]{x} - 1.$$

we give the following reformulation of the cancellation equations as above using x as the independent variable for both f and f^{-1}.

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in the domain of } f$$
$$f(f^{-1}(x)) = x \quad \text{for every } x \text{ in the domain of } f^{-1}$$

DOMAIN AND RANGE OF INVERSE FUNCTIONS: The equations in (3) imply the following relationships between the domains and ranges of f and f^{-1}:

$$\text{domain of } f^{-1} = \text{range of } f$$
$$\text{range of } f^{-1} = \text{domain of } f$$



▲ **Figure 0.4.2** If f maps a to b, then f^{-1} maps b back to a.

One way to show that two sets are the same is to show that each is a subset of the other.

0.4.2 THEOREM *If an equation $y = f(x)$ can be solved for x as a function of y, say $x = g(y)$, then f has an inverse and that inverse is $g(y) = f^{-1}(y)$.*

A Procedure for Finding the Inverse of a Function f

Step 1. Write down the equation $y = f(x)$.

Step 2. If possible, solve this equation for x as a function of y.

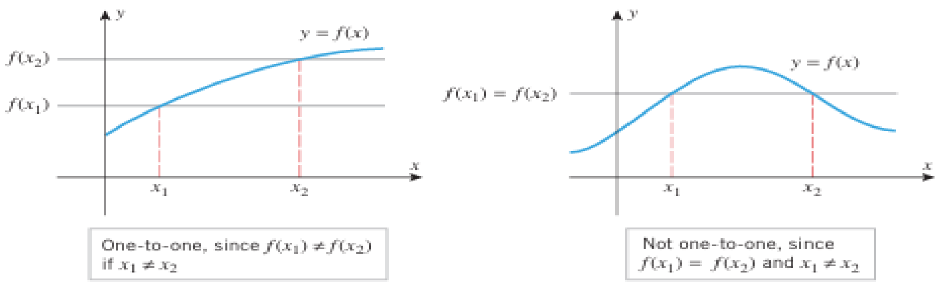
Step 3. The resulting equation will be $x = f^{-1}(y)$, which provides a formula for f^{-1} with y as the independent variable.

Step 4. If y is acceptable as the independent variable for the inverse function, then you are done, but if you want to have x as the independent variable, then you need to interchange x and y in the equation $x = f^{-1}(y)$ to obtain $y = f^{-1}(x)$.

EXISTENCE OF INVERSE FUNCTIONS:

0.4.3 THEOREM *A function has an inverse if and only if it is one-to-one.*

Stated algebraically, a function f is one-to-one if and only if $f(x_1) = f(x_2)$ whenever $x_1 = x_2$; stated geometrically, a function f is one-to-one if and only if the graph of $y = f(x)$ is cut at most once by any horizontal line. The latter statement together with Theorem provides the following geometric test for determining whether a function has an inverse.



0.4.4 THEOREM ***(The Horizontal Line Test)*** *A function has an inverse function if and only if its graph is cut at most once by any horizontal line.*

INCREASING ORDECREASING FUNCTIONS ARE INVERTIBLE:

A function whose graph is always rising as it is traversed from left to right is said to be an increasing function, and a function whose graph is always falling as it is traversed from left to right is said to be decreasing function. If x1 and x2 are points in the domain of a function f,

then f is increasing if $f(x_1) < f(x_2)$ where $x_1 < x_2$

and f is decreasing if $f(x_1) > f(x_2)$ where $x_1 < x_2$

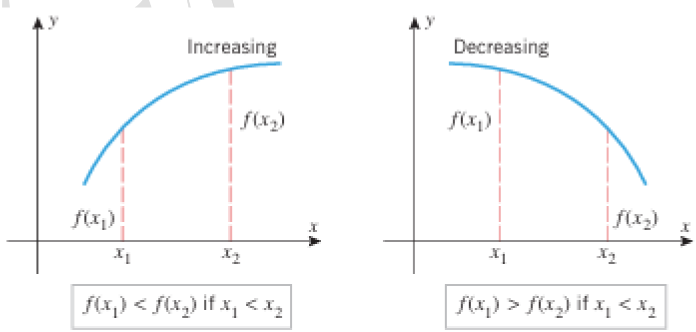
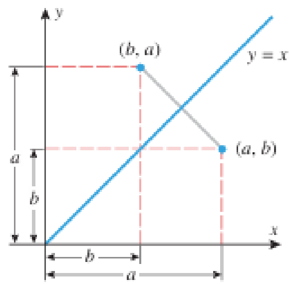


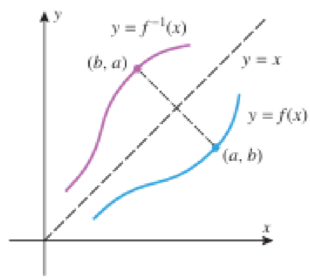
Figure 0.4.7

GRAPHS OF INVERSE FUNCTIONS:

0.4.5 THEOREM *If f has an inverse, then the graphs of $y = f(x)$ and $y = f^{-1}(x)$ are reflections of one another about the line $y = x$; that is, each graph is the mirror image of the other with respect to that line.*

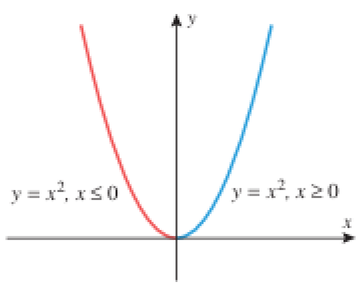


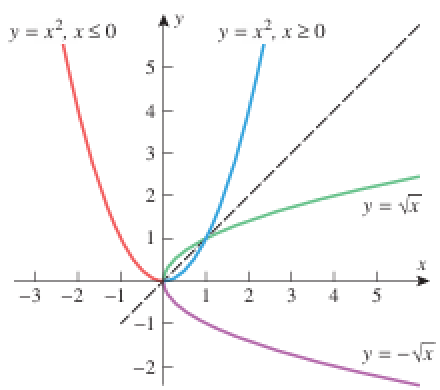
The points (a, b) and (b, a) are reflections about $y = x$.



The graphs of f and f^{-1} are reflections about $y = x$.

RESTRICTING DOMAINS FOR INVERTIBILITY: If a function g is obtained from a function f by placing restrictions on the domain of f, then g is called a restriction of f. Thus, for example, the function $g(x) = x^3$, $x \geq 0$ is a restriction of the function $f(x)= x^3$. More precisely, it is called the restriction of x^3 to the interval [0,+). Sometimes it is possible to create an invertible function from a function that is not invertible by restricting the domain appropriately. For example, we showed earlier that $f(x)= x^2$ is not invertible. However, consider the restricted functions $f_1(x) = x^2$, $x \geq 0$ and $f_2(x)=x^2$, $x \leq 0$ the union of whose graphs is the complete graph of $f(x)= x^2$. These restricted functions are each one-to-one (hence invertible), since their graphs pass the horizontal line test. As illustrated in Figure, their inverses are $f^{-1}{}_1 (x) =\sqrt{x}$ and $f^{-1}{}_2 (x) =-\sqrt{x}$.





INVERSE FRIGONOMETRIC FUNCTIONS:

The six basic trigonometric functions do not have inverses because their graphs repeat periodically and hence do not pass the horizontal line test. To come over this problem we will restrict the domains of the trigonometric functions to produce one to one functions to define the inverse trigonometric functions:

The following formal definitions summarize the preceding discussion.

0.4.6 DEFINITION The ***inverse sine function***, denoted by $\sin^{-1}$, is defined to be the inverse of the restricted sine function

$$\sin x, \quad -\pi/2 \le x \le \pi/2$$

0.4.7 DEFINITION The ***inverse cosine function***, denoted by $\cos^{-1}$, is defined to be the inverse of the restricted cosine function

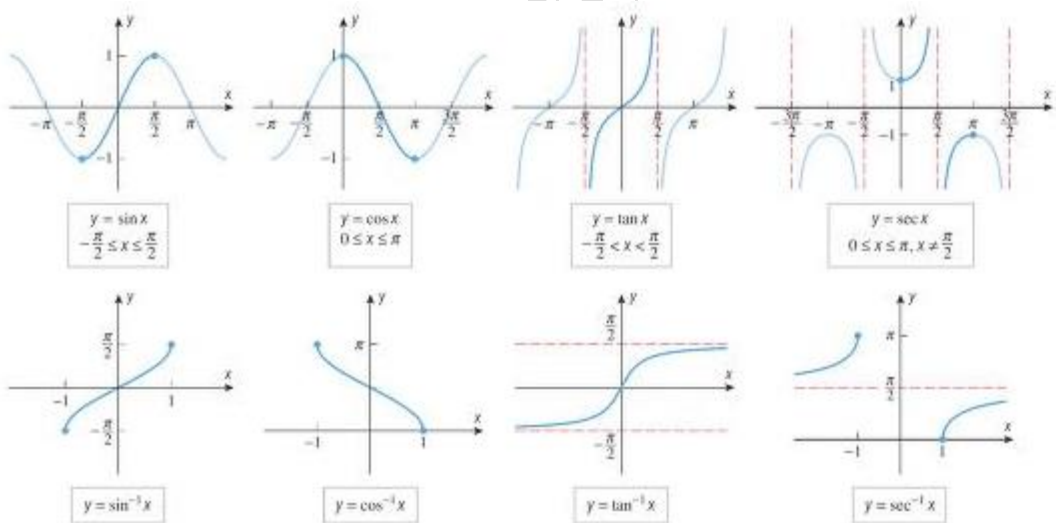
$$\cos x, \quad 0 \le x \le \pi$$

0.4.8 DEFINITION The ***inverse tangent function***, denoted by $\tan^{-1}$, is defined to be the inverse of the restricted tangent function

$$\tan x, \quad -\pi/2 < x < \pi/2$$

0.4.9 DEFINITION* The ***inverse secant function***, denoted by $\sec^{-1}$, is defined to be the inverse of the restricted secant function

$$\sec x, \quad 0 \le x \le \pi \text{ with } x \ne \pi/2$$



PROPERTIES OF INVERSE TRIGONOMETRIC FUNCTIONS

FUNCTION	DOMAIN	RANGE	BASIC RELATIONSHIPS
$\sin^{-1}$	$[-1, 1]$	$[-\pi/2, \pi/2]$	$\sin^{-1}(\sin x) = x$ if $-\pi/2 \le x \le \pi/2$ $\sin(\sin^{-1} x) = x$ if $-1 \le x \le 1$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$	$\cos^{-1}(\cos x) = x$ if $0 \le x \le \pi$ $\cos(\cos^{-1} x) = x$ if $-1 \le x \le 1$
$\tan^{-1}$	$(-\infty, +\infty)$	$(-\pi/2, \pi/2)$	$\tan^{-1}(\tan x) = x$ if $-\pi/2 < x < \pi/2$ $\tan(\tan^{-1} x) = x$ if $-\infty < x < +\infty$
$\sec^{-1}$	$(-\infty, -1] \cup [1, +\infty)$	$[0, \pi/2) \cup (\pi/2, \pi]$	$\sec^{-1}(\sec x) = x$ if $0 \le x \le \pi, x \ne \pi/2$ $\sec(\sec^{-1} x) = x$ if $\lvert x\rvert \ge 1$

Note: A reflection about y=x converts vertical line into horizontal lines, and vice versa, and it converts x-intercepts into y-intercept and vice versa.

Exponential and Logarithmic Function:

Irrational Exponents: If b is a nonzero real number then nonzero integer powers of b are defined by $b^n = b * b * b \dots . * b$ and $b^{-n} = {}^{1}/_{b^n}$

And if n=0 then $b^0 = 1$, also if ${}^{p}/_{q}$ is a +ve rational number expressed in lowest terms, then

$b^{\frac{p}{q}} = \sqrt[q]{b^p} = (\sqrt[q]{b}\,)^{p}$ and $b^{-\frac{p}{q}} = {}^{1}/_{b^{p/q}}$ Here we assume b>0

If b is –ve, then some fractional powers of b will have imaginary values such as 2^{π}, $3^{\sqrt{2}}$, π^{-7}

Laws of Exponent:

$i.\ b^p b^q = b^{p+q}$ ii. $b^p / b^q = b^{p-q}$ iii. $(b^p)^q\ b^{pq}$ iv. $b^0 = 1$ **v.** $b^{\frac{p}{q}} = \sqrt[q]{b^p} = (\sqrt[q]{b}\,)^{p}$

vi. $b^{-\frac{p}{q}} = {}^{1}/_{b^{p/q}}$

Range of exponential function is (0,+∞) Domain of exponential function is (-∞,+∞)

The Family Of Exponent Functions: a function of the form f(x) = b^x,

where b > 0 is called an exponential function with base b. for e.g. f(x) = 2^x, $(1/2)^x$, π^x.

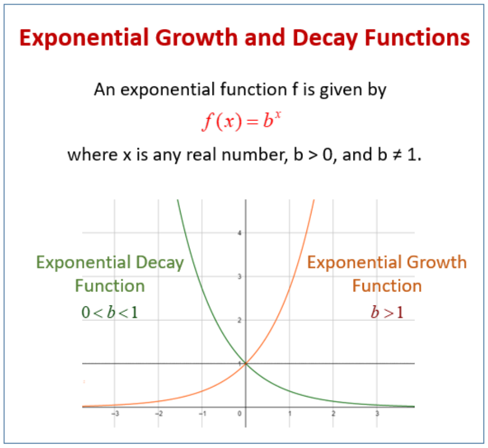
An exponential function has a constant base and variable exponent.

The graph of y = b^x has the following properties:-

- The graph passes through (0,1) because $b^0 = 1$
- If b > 1, the value of b^x increases as x increase. As you traverse the graph from left to right, the value of b^x increases indefinitely .
- If you traverse the graph from right to left, the value of b^x decreases toward zero but never reach zero .Thus, the x-axis is a horizontal asymptote of the graph of b^x.
- If 0 < b < 1, the value of b^x decreases as x increases and vice versa of point 2.
- If b = 1, then the value of b^x is constant.

The graph of $y = (1/\,b)^x$ is the reflection of the graph of $y = b^x$ about the y-axis because of replacing b by –x in the equation

$$y = b^x \Rightarrow y = b^{-x} \Rightarrow y = (1/\,b)^x$$



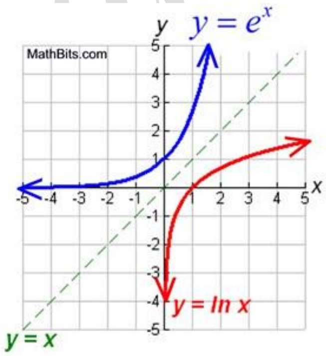
The Natural Exponential Function:

The function f(x)= e^x is called the natural exponential function. It is also denoted as exp(x), in which case the relationship

$$e^{x_1} + e^{x_2} = e^{x_1}\, e^{x_2}$$

That base, denoted by the letter e, is a certain irrational number whose value to six decimal place is e ≈ 2.718282.

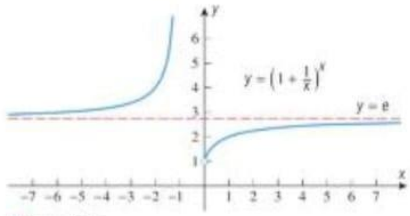
Also, b=e is the only base for which the slope of the tangent line to the curve $y = b^x$ at any point P on the curve is equal to y-coordinate at P. Thus, the tangent line to $y = e^x$ at (0,1) has slope 1.



The constant e also rises in the context of the graph of the equation.

$$y = \left(1 + \frac{1}{x}\right)^x$$

Y=e is a horizontal asymptote of this graph. As a result, the value of e can be approximated to any degree of accuracy by evaluating for x sufficiently large in absolute value.



▲ Figure 0.5.5

Table 0.5.2

APPROXIMATIONS OF e BY $(1 + 1/x)^x$ FOR INCREASING VALUES OF x

x	$1 + \frac{1}{x}$	$\left(1 + \frac{1}{x}\right)^x$
1	2	≈ 2.000000
10	1.1	2.593742
100	1.01	2.704814
1000	1.001	2.716924
10,000	1.0001	2.718146
100,000	1.00001	2.718268
1,000,000	1.000001	2.718280

Change of Base Formula for Logarithms:

Y = log_b x ----- i x = b^y ----- ii

Taking logarithm on both sides of eqn. (ii)

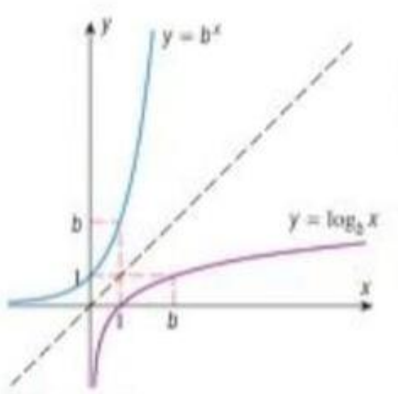
ylnb = lnx

=>y = lnx/lnb

=> log_b x = lnx/lnb

Logarithmic functions: The function f(x)=log_b x the logarithmic function with base b. Logarithmic functions can also be viewed as inverses of exponential functions. if b > 1 and b ≠ 1, then the graph of f(x) = b^x passes the horizontal line test. So, b^x has an inverse formula for this inverse with x as the independent variable is x = b^y for y as a function of x and can be written as y=log_b x.

Theorem:- If b > 1 and b ≠ 1, then b^x and y=log_b x are reflections of one another about the line y=x. In general; y = lnx iff x = e^y



Log Formula

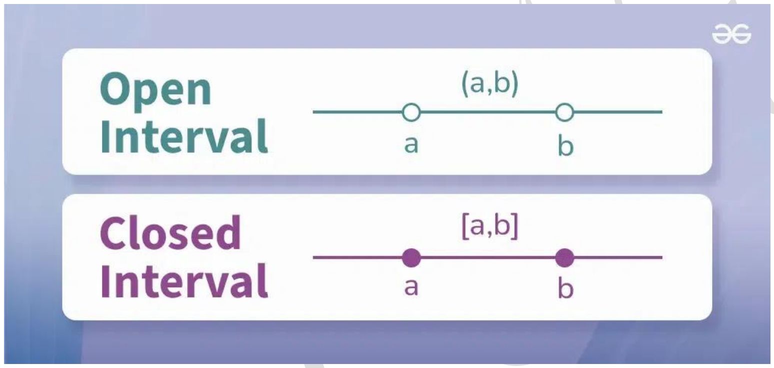
$$\log_b(y) = x$$
$$\log_b(xy) = \log_b(x) + \log_b(y)$$
$$\log_b(\tfrac{x}{y}) = \log_b(x) - \log_b(y)$$
$$\log_b(x^d) = d\log_b(x)$$
$$\log_b\left(\sqrt[y]{x}\right) = \frac{\log_b(x)}{y}$$
$$\log_b x = \frac{\log_k x}{\log_k b}$$

Range of logarithmic function is (-∞,+∞) Domain of logarithmic function is (0,+∞)

• **Open intervals**
An open interval is a set of numbers that doesn't include its endpoints,
Use parentheses to indicate that the endpoints are not included. For example, (a, b) is an open interval that includes all real numbers between a and b but not a & b.

• **Closed intervals**
while a closed interval is a set of numbers that does include its endpoints. Use square brackets to indicate that the endpoints are included. For example, [a, b] open bracket a comma b close bracket $[a, b]$ is a closed interval that includes all real numbers between a and b, including a and b.

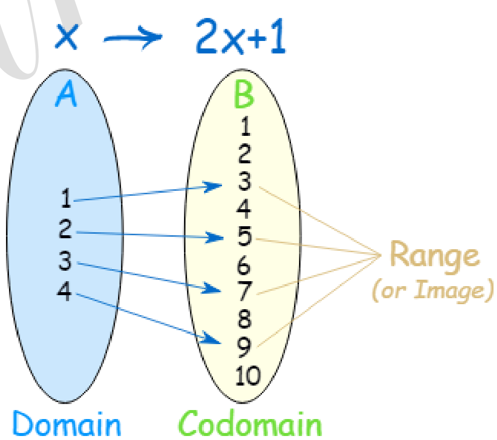
• **Half-open intervals**
Include one endpoint but not the other. They can be left-open or right-open, depending on which endpoint is excluded.



Domain: refers to all possible input values of a function.

Codomain: refers to the set of all possible output values that a function could produce,

Range: refers to the set of actual output values that the function generates when applied to its domain.

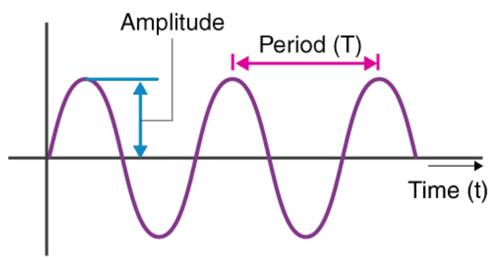


Difference Between Symmetry And Reflection: Symmetry is when a shape or object is the same on both sides, while reflection is when an object is reflected across a line to create a mirror image.

Amplitude: It is the maximum displacement or distance moved by a point on a vibrating body or wave measured from its equilibrium position. i.e. A= $|A|$

Frequency: The rate at which something occurs or is repeated over a particular period of time (usually 1 sec) or in a given sample. F = $|B|/_{2\pi}$

Period: It is the time it takes for a vibrating object to complete one cycle of oscillation or vibration. P = $2\pi/_{|B|}$



Exercise 0. 1

1. Use the accompanying graph to answer the following questions, making reasonable approximations where needed.
(a) For what values of x is $y = 1$?
(b) For what values of x is $y = 3$?
(c) For what values of y is $x = 3$?
(d) For what values of x is $y \leq 0$?
(e) What are the maximum and minimum values of y and for what values of x do they occur?

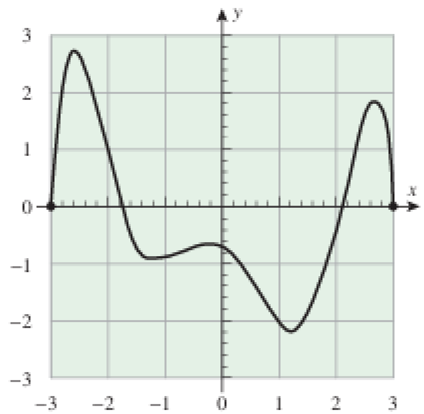


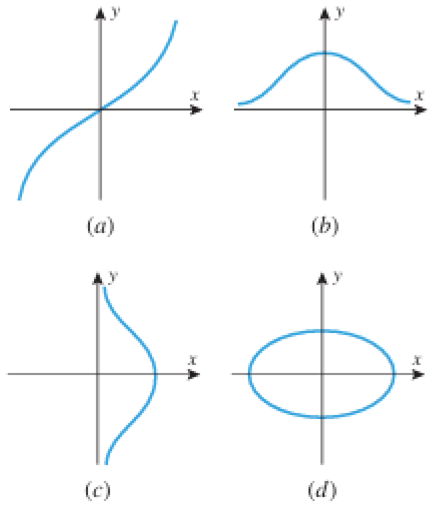
Figure Ex-1

2. Use the accompanying table to answer the questions posed in Exercise 1.

x	−2	−1	0	2	3	4	5	6
y	5	1	−2	7	−1	1	0	9

Table Ex-2

3. In each part of the accompanying figure, determine whether the graph defines y as a function of x.



▲ **Figure Ex-3**

4. In each part, compare the natural domains of f and g.

(a) $f(x) = \dfrac{x^2 + x}{x + 1}$; $g(x) = x$

(b) $f(x) = \dfrac{x\sqrt{x} + \sqrt{x}}{x + 1}$; $g(x) = \sqrt{x}$

7. Find $f(0)$, $f(2)$, $f(-2)$, $f(3)$, $f(\sqrt{2})$, and $f(3t)$.

(a) $f(x) = 3x^2 - 2$ (b) $f(x) = \begin{cases} \dfrac{1}{x}, & x > 3 \\ 2x, & x \le 3 \end{cases}$

8. Find $g(3)$, $g(-1)$, $g(\pi)$, $g(-1.1)$, and $g(t^2 - 1)$.

(a) $g(x) = \dfrac{x + 1}{x - 1}$ (b) $g(x) = \begin{cases} \sqrt{x + 1}, & x \ge 1 \\ 3, & x < 1 \end{cases}$

9–10 Find the natural domain and determine the range of each :

9. (a) $f(x) = \dfrac{1}{x-3}$ (b) $F(x) = \dfrac{x}{|x|}$

(c) $g(x) = \sqrt{x^2 - 3}$ (d) $G(x) = \sqrt{x^2 - 2x + 5}$

(e) $h(x) = \dfrac{1}{1 - \sin x}$ (f) $H(x) = \sqrt{\dfrac{x^2-4}{x-2}}$

10. (a) $f(x) = \sqrt{3 - x}$ (b) $F(x) = \sqrt{4 - x^2}$

(c) $g(x) = 3 + \sqrt{x}$ (d) $G(x) = x^3 + 2$

(e) $h(x) = 3\sin x$ (f) $H(x) = (\sin\sqrt{x})^{-2}$

19–22 True–False Determine whether the statement is true or false. Explain your answer. ■

19. A curve that crosses the x-axis at two different points cannot be the graph of a function.

20. The natural domain of a real-valued function defined by a formula consists of all those real numbers for which the formula yields a real value.

21. The range of the absolute value function is all positive real numbers.

22. If $g(x) = 1/\sqrt{f(x)}$, then the domain of g consists of all those real numbers x for which $f(x) \neq 0$.

23. Use the equation $y = x^2 - 6x + 8$ to answer the following questions.

(a) For what values of x is $y = 0$?

(b) For what values of x is $y = -10$?

(c) For what values of x is $y \geq 0$?

(d) Does y have a minimum value? A maximum value? If so, find them.

24. Use the equation $y = 1 + \sqrt{x}$ to answer the following questions.

(a) For what values of x is $y = 4$?

(b) For what values of x is $y = 0$?

(c) For what values of x is $y \geq 6$?

(cont.)

(d) Does y have a minimum value? A maximum value? If so, find them.

Exercise 0.2

5–24 Sketch the graph of the equation by translating, reflecting, compressing, and stretching the graph of $y = x^2$, $y = \sqrt{x}$, $y = 1/x$, $y = |x|$, or $y = \sqrt[3]{x}$ appropriately. Then use a graphing utility to confirm that your sketch is correct. ■

5. $y = -2(x+1)^2 - 3$

6. $y = \frac{1}{2}(x-3)^2 + 2$

7. $y = x^2 + 6x$

8. $y = \frac{1}{2}(x^2 - 2x + 3)$

9. $y = 3 - \sqrt{x+1}$

10. $y = 1 + \sqrt{x-4}$

11. $y = \frac{1}{2}\sqrt{x} + 1$

12. $y = -\sqrt{3x}$

13. $y = \dfrac{1}{x-3}$

14. $y = \dfrac{1}{1-x}$

15. $y = 2 - \dfrac{1}{x+1}$

16. $y = \dfrac{x-1}{x}$

17. $y = |x+2| - 2$

18. $y = 1 - |x-3|$

19. $y = |2x-1| + 1$

20. $y = \sqrt{x^2 - 4x + 4}$

21. $y = 1 - 2\sqrt[3]{x}$

22. $y = \sqrt[3]{x-2} - 3$

23. $y = 2 + \sqrt[3]{x+1}$

24. $y + \sqrt[3]{x-2} = 0$

27–28 Find formulas for $f+g$, $f-g$, fg, and f/g, and state the domains of the functions. ■

27. $f(x) = 2\sqrt{x-1}$, $g(x) = \sqrt{x-1}$

28. $f(x) = \dfrac{x}{1+x^2}$, $g(x) = \dfrac{1}{x}$

29. Let $f(x) = \sqrt{x}$ and $g(x) = x^3 + 1$. Find

(a) $f(g(2))$ (b) $g(f(4))$ (c) $f(f(16))$

(d) $g(g(0))$ (e) $f(2+h)$ (f) $g(3+h)$.

30. Let $g(x) = \sqrt{x}$. Find

(a) $g(5s+2)$ (b) $g(\sqrt{x}+2)$ (c) $3g(5x)$

(d) $\dfrac{1}{g(x)}$ (e) $g(g(x))$ (f) $(g(x))^2 - g(x^2)$

(g) $g(1/\sqrt{x})$ (h) $g((x-1)^2)$ (i) $g(x+h)$.

31–34 Find formulas for $f \circ g$ and $g \circ f$, and state the domains of the compositions. ■

31. $f(x) = x^2$, $g(x) = \sqrt{1-x}$

32. $f(x) = \sqrt{x-3}$, $g(x) = \sqrt{x^2+3}$

33. $f(x) = \dfrac{1+x}{1-x}$, $g(x) = \dfrac{x}{1-x}$

34. $f(x) = \dfrac{x}{1+x^2}$, $g(x) = \dfrac{1}{x}$

35–36 Find a formula for $f \circ g \circ h$. ■

35. $f(x) = x^2 + 1$, $g(x) = \dfrac{1}{x}$, $h(x) = x^3$

36. $f(x) = \dfrac{1}{1+x}$, $g(x) = \sqrt[3]{x}$, $h(x) = \dfrac{1}{x^3}$

37–42 Express f as a composition of two functions; that is, find g and h such that $f = g \circ h$. [*Note:* Each exercise has more than one solution.] ■

37. (a) $f(x) = \sqrt{x+2}$ (b) $f(x) = |x^2 - 3x + 5|$

38. (a) $f(x) = x^2 + 1$ (b) $f(x) = \dfrac{1}{x-3}$

39. (a) $f(x) = \sin^2 x$ (b) $f(x) = \dfrac{3}{5 + \cos x}$

40. (a) $f(x) = 3\sin(x^2)$ (b) $f(x) = 3\sin^2 x + 4\sin x$

41. (a) $f(x) = \left(1 + \sin(x^2)\right)^3$ (b) $f(x) = \sqrt{1 - \sqrt[3]{x}}$

42. (a) $f(x) = \dfrac{1}{1-x^2}$ (b) $f(x) = |5 + 2x|$

43–46 True–False Determine whether the statement is true or false. Explain your answer. ■

43. The domain of $f + g$ is the intersection of the domains of f and g.

44. The domain of $f \circ g$ consists of all values of x in the domain of g for which $g(x) \neq 0$.

45. The graph of an even function is symmetric about the y-axis.

46. The graph of $y = f(x+2) + 3$ is obtained by translating the graph of $y = f(x)$ right 2 units and up 3 units.

Exercise 0.3

1. (a) Find an equation for the family of lines whose members have slope $m = 3$.
(b) Find an equation for the member of the family that passes through $(-1, 3)$.
(c) Sketch some members of the family, and label them with their equations. Include the line in part (b).

2. Find an equation for the family of lines whose members are perpendicular to those in Exercise 1.

3. (a) Find an equation for the family of lines with y-intercept $b = 2$.
(b) Find an equation for the member of the family whose angle of inclination is $135°$.
(c) Sketch some members of the family, and label them with their equations. Include the line in part (b).

4. Find an equation for
(a) the family of lines that pass through the origin
(b) the family of lines with x-intercept $a = 1$
(c) the family of lines that pass through the point $(1, -2)$
(d) the family of lines parallel to $2x + 4y = 1$.

5. Find an equation for the family of lines tangent to the circle with center at the origin and radius 3.

6. Find an equation for the family of lines that pass through the intersection of $5x - 3y + 11 = 0$ and $2x - 9y + 7 = 0$.

8. Find all lines through $(6, -1)$ for which the product of the x- and y-intercepts is 3.

12. The accompanying table gives approximate values of three functions: one of the form kx^2, one of the form kx^{-3}, and one of the form $kx^{3/2}$. Identify which is which, and estimate k in each case.

x	0.25	0.37	2.1	4.0	5.8	6.2	7.9	9.3
$f(x)$	640	197	1.08	0.156	0.0513	0.0420	0.0203	0.0124
$g(x)$	0.0312	0.0684	2.20	8.00	16.8	19.2	31.2	43.2
$h(x)$	0.250	0.450	6.09	16.0	27.9	30.9	44.4	56.7

▲ **Table Ex-12**

35–36 Find the amplitude and period, and sketch at least two periods of the graph by hand. If you have a graphing utility, use it to check your work.

35. (a) $y = 3\sin 4x$ (b) $y = -2\cos \pi x$
(c) $y = 2 + \cos\left(\dfrac{x}{2}\right)$

36. (a) $y = -1 - 4\sin 2x$ (b) $y = \frac{1}{2}\cos(3x - \pi)$
(c) $y = -4\sin\left(\dfrac{x}{3} + 2\pi\right)$

Exercise 0.4

1. In (a)–(d), determine whether f and g are inverse functions.
(a) $f(x) = 4x,\ g(x) = \frac{1}{4}x$
(b) $f(x) = 3x + 1,\ g(x) = 3x - 1$
(c) $f(x) = \sqrt[3]{x - 2},\ g(x) = x^3 + 2$
(d) $f(x) = x^4,\ g(x) = \sqrt[4]{x}$

3. In each part, use the horizontal line test to determine whether the function f is one-to-one.
(a) $f(x) = 3x + 2$
(b) $f(x) = \sqrt{x - 1}$
(c) $f(x) = |x|$
(d) $f(x) = x^3$
(e) $f(x) = x^2 - 2x + 2$
(f) $f(x) = \sin x$

9–16 Find a formula for $f^{-1}(x)$.

9. $f(x) = 7x - 6$

10. $f(x) = \dfrac{x+1}{x-1}$

11. $f(x) = 3x^3 - 5$

12. $f(x) = \sqrt[5]{4x + 2}$

13. $f(x) = 3/x^2,\quad x < 0$

14. $f(x) = 5/(x^2 + 1),\quad x \geq 0$

15. $f(x) = \begin{cases} 5/2 - x, & x < 2 \\ 1/x, & x \geq 2 \end{cases}$

16. $f(x) = \begin{cases} 2x, & x \leq 0 \\ x^2, & x > 0 \end{cases}$

17–20 Find a formula for $f^{-1}(x)$, and state the domain of the function f^{-1}.

17. $f(x) = (x + 2)^4,\quad x \geq 0$

18. $f(x) = \sqrt{x + 3}$

19. $f(x) = -\sqrt{3 - 2x}$

20. $f(x) = x - 5x^2,\quad x \geq 1$

31. Let $f(x) = 2x^3 + 5x + 3$. Find x if $f^{-1}(x) = 1$.

32. Let $f(x) = \dfrac{x^3}{x^2 + 1}$. Find x if $f^{-1}(x) = 2$.

34. (a) Prove: If f and g are one-to-one, then so is the composition $f \circ g$.
(b) Prove: If f and g are one-to-one, then

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$$

Exercise 0.5

5–6 Find the exact value of the expression without using a calculating utility.

5. (a) $\log_2 16$ (b) $\log_2\left(\frac{1}{32}\right)$
(c) $\log_4 4$ (d) $\log_9 3$

6. (a) $\log_{10}(0.001)$ (b) $\log_{10}(10^4)$
(c) $\ln(e^3)$ (d) $\ln(\sqrt{e})$

9–10 Use the logarithm properties in Theorem 0.5.2 to rewrite the expression in terms of r, s, and t, where $r = \ln a$, $s = \ln b$, and $t = \ln c$.

9. (a) $\ln a^2\sqrt{bc}$ (b) $\ln \dfrac{b}{a^3c}$

10. (a) $\ln \dfrac{\sqrt[3]{c}}{ab}$ (b) $\ln \sqrt{\dfrac{ab^3}{c^2}}$

11–12 Expand the logarithm in terms of sums, differences, and multiples of simpler logarithms.

11. (a) $\log(10x\sqrt{x-3})$ (b) $\ln \dfrac{x^2 \sin^3 x}{\sqrt{x^2+1}}$

12. (a) $\log \dfrac{\sqrt[3]{x+2}}{\cos 5x}$ (b) $\ln \sqrt{\dfrac{x^2+1}{x^3+5}}$

13–15 Rewrite the expression as a single logarithm.

13. $4\log 2 - \log 3 + \log 16$

14. $\frac{1}{2}\log x - 3\log(\sin 2x) + 2$

15. $2\ln(x+1) + \frac{1}{3}\ln x - \ln(\cos x)$

16–23 Solve for x without using a calculating utility.

16. $\log_{10}(1+x) = 3$

17. $\log_{10}(\sqrt{x}) = -1$

18. $\ln(x^2) = 4$

19. $\ln(1/x) = -2$

20. $\log_3(3^x) = 7$

21. $\log_5(5^{2x}) = 8$

22. $\ln 4x - 3\ln(x^2) = \ln 2$

23. $\ln(1/x) + \ln(2x^3) = \ln 3$

24–29 Solve for x without using a calculating utility. Use the natural logarithm anywhere that logarithms are needed.

24. $3^x = 2$

25. $5^{-2x} = 3$

26. $3e^{-2x} = 5$

27. $2e^{3x} = 7$

28. $e^x - 2xe^x = 0$

29. $xe^{-x} + 2e^{-x} = 0$

30. Solve $e^{-2x} - 3e^{-x} = -2$ for x without using a calculating utility. [*Hint:* Rewrite the equation as a quadratic equation in $u = e^{-x}$.]